Student Record Management System Using Python

1. Introduction

In schools and colleges, keeping student records by hand can take a lot of time and may lead to mistakes. Using a computer system makes the work faster and more accurate. This project is about creating a Student Record Management System using Python to store and manage student information easily.

2. Problem Statement

Managing student records manually can cause problems like losing data, repeating information, or taking too long to find details. So, there is a need for a simple system that can store and manage student data in an organized way.

3. Objectives

- To create a system to manage student records
- To store student information using file handling
- To search, update, and delete student records
- To make data management easier and faster

4. Methodology

This system is developed using the Python programming language. It uses functions and file handling to save and read data from a JSON file. The program has a menu that allows the user to choose different options like adding, viewing, searching, updating, and deleting student records.

5. Tools and Technologies Used

- Python
- JSON File
- Visual Studio Code
- Windows Operating System

6. Implementation

The program is divided into different functions, such as:

- Add Student
- View Students
- Search Student
- Update Student
- Delete Student

All student data is saved in a file called students.json.

7. Results

The system works properly and performs all the required tasks for managing student records. It provides a simple and efficient way to store and handle student information.

8. Challenges Faced

- Learning how file handling works in Python
- Storing data correctly
- Managing user input without errors

9. Conclusion

This project shows how Python can be used to solve real-life problems. It helps in managing student records more easily and reduces mistakes compared to manual work.

10. Future Scope

- Add a graphical user interface (GUI)
- Use a database like MySQL instead of a file
- Add a login system for security